

Asymptotic convergence of heterogeneous first-order aggregation models: from the sphere to the unitary group

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High-dimensional Kuramoto models

- ▶ Kuramoto model [Kuramoto '75]:

$$\dot{\theta}_j = \nu_j + \frac{\kappa}{N} \sum_{k=1}^N \sin(\theta_k - \theta_j) \quad (1)$$

- $\theta_j = \theta_j(t)$: oscillator at j -th node
 - ν_j : intrinsic natural frequency acted on j -th node
 - κ : uniform coupling strength between oscillators
- ▶ Question: Can the Kuramoto model be extended to high-dimensional one?
 - ▶ We notice that the Kuramoto model is defined on $\mathbf{U}(1)$ or $\mathbb{S}^1$.

High-dimensional Kuramoto on unitary group

- ▶ High-dimensional Kuramoto model on unitary group has been proposed in [Lohe J. Phys. A '09]:

$$\dot{U}_j = -iH_j U_j + \frac{\kappa}{2N} \sum_{k=1}^N (U_k - U_k U_j^\dagger U_j) \quad (2)$$

- $U_j = U_j(t) \in \mathbf{U}(d)$: oscillator at j -th node
 - H_j : $d \times d$ Hermitian matrix at j -th node which plays a role of natural frequency
- ▶ Unitarity is preserved.
 - ▶ The Kuramoto model is (formally) derived from (2) by the one-dimensional ansatz:

$$U_j = e^{-i\theta_j}, \quad H_j = \nu_j$$

- ▶ Question: Does this model also exhibit emergent behavior similar to the Kuramoto model?

Identical H_j

- As expected from Kuramoto literature, if H_j 's are identical, then one can show that relative distances between U_j converge to zero.

Theorem (Ha-Ryoo J. Stat. Phys '16)

Let $\{U_j\}$ be a solution to (2) with $\kappa > 0$ and $H_j \equiv H$. If $D(U_0) < \sqrt{2}$, then we have

$$\lim_{t \rightarrow \infty} D(U(t)) = 0.$$

Here, $D(U(t))$ is defined as the maximal diameter:

$$D(U(t)) := \max_{1 \leq i, j \leq N} \|U_i(t) - U_j(t)\|.$$

In fact, when $H_j \equiv H$, we would set $H = O_d$ and then system (2) with $H_j \equiv O_d$ can be represented as a gradient flow. Hence, a solution converges to equilibrium. Hence, we know

1. Convergence of U_j toward U_j^∞
2. Structure of $\{U_j^\infty\}$

- Question: Then, what happens for distinct H_j ?

Theorem (Ha-Ryoo J. Stat. Phys. '16)

Let $\{U_j\}$ be a solution to (2) with $\kappa > 0$ and $H_i \neq H_j$ in general. If $\kappa \gg 1$ and $D(U_0) \ll 1$, then the limit of $U_i U_j^\dagger$ exists.

Note that

$$\|U_i - U_j\|^2 = 2\text{Re}(I_d - U_i U_j^\dagger).$$

Question: Do we know that the limit of U_i itself, say, U_i^∞ , exists? If yes, is there any specific structure for $\{U_i^\infty\}$?

- ▶ Recall that when $H_j \equiv H$, a solution converges to equilibrium.
- ▶ However when $H_i \neq H_j$, the system cannot be represented as a gradient flow, and is considered as (small) perturbation of a gradient flow.
- ▶ Hence, we do not know whether U_j converges to equilibrium or not.

Theorem (K.-Park, arXiv:2206.00984)

Suppose a priori that the limit of $U_i U_j^\dagger$ exists, and further assume that

$$\sum_{j=1}^N H_j = O_d.$$

Then, there exists U_j^∞ such that

$$\lim_{t \rightarrow \infty} U_j(t) = U_j^\infty.$$

Question: Is there any specific structure for the equilibrium $\{U_j^\infty\}$?

- We assume

$$H_j = a_j I_d, \quad \sum_{j=1}^N a_j = 0$$

so that degree of freedom is just 1. In this case, the system reduces to

$$\dot{U}_j = i a_j U_j + \frac{\kappa}{2N} \sum_{k=1}^N (U_k - U_j U_k^\dagger U_j).$$

- We recall that from the ansatz $U_j = e^{i\theta_j}$, system (2) reduces to the Kuramoto model:

$$\dot{\theta}_j = a_j + \frac{\kappa}{N} \sum_{k=1}^N \sin(\theta_k - \theta_j).$$

- We assume zero-sum condition:

$$\sum_{j=1}^N a_j = 0.$$

- Fortunately, there are many results for Kuramoto synchronization.

Theorem (Ha-Ryoo Comm. Math. Phys. '20)

Let $\{\theta_j\}$ be a solution to (1) with $\sum_{j=1}^N \nu_j = 0$. Then, there exists a critical coupling strength $\kappa_c > 0$ such that if $\kappa > \kappa_c$, then the limit of θ_i exists.

- We define an auxiliary variable $V_j \in \mathbf{U}(d)$:

$$V_j(t) := e^{-i\theta_j(t)} U_j(t).$$

- Then, V_j satisfies

$$\dot{V}_j = \frac{\kappa}{2N} \sum_{k=1}^N \left[e^{i(\theta_k - \theta_j)} (V_k - V_j) - e^{i(\theta_j - \theta_k)} V_j (V_k^\dagger - V_j^\dagger) V_j \right] \quad (3)$$

which looks like ‘identical’.

- In fact, heterogeneity (induced from natural frequency term a_j) is absorbed in θ_j .
- Then, one can show that if $D(V_0) \ll 1$ and $D(\Theta_0) \ll 1$, then there exists $V^\infty \in \mathbf{U}(d)$ such that

$$\lim_{t \rightarrow \infty} V_j(t) = V^\infty.$$

Here,

$$D(V(t)) := \max_{1 \leq i, j \leq N} \|V_i(t) - V_j(t)\|, \quad D(\Theta(t)) := \max_{1 \leq i, j \leq N} \|\theta_i(t) - \theta_j(t)\|.$$

► To this end, since we know $\theta_j(t) \rightarrow \theta_j^\infty$ and $V_j(t) \rightarrow V^\infty$,

$$U_j = V_j(t)e^{i\theta_j(t)} \rightarrow e^{i\theta_j^\infty} V^\infty.$$

Theorem (K.-Park, arXiv:2206.00984)

Let $\{U_j\}$ be a solution to (2) with $H_j = a_j I_d$. If $\kappa \gg 1$ and $D(U_0) \ll 1$, then there exist $\theta_j^\infty \in \mathbb{R}$ and $V^\infty \in U(d)$ such that

$$\lim_{t \rightarrow \infty} U_j(t) = e^{i\theta_j^\infty} V^\infty =: U_j^\infty.$$

Hence, we know

1. Convergence of U_j toward U_j^∞
2. Structure of $\{U_j^\infty\}$

Question: So far, we assume that there is only one parameter for $H_j = a_j I_d$. Can we increase degree of freedom?

- In fact, our method can be extended to diagonal case:

$$H_j = E_j = \text{diag}(e_1, \dots, e_d) : \text{diagonal}, \quad \sum_{j=1}^N E_j = O_d.$$

- Then, we associate matrix-valued Kuramoto model:

$$\dot{D}_j = E_j + \frac{\kappa}{N} \sum_{k=1}^N \sin(D_k - D_j)$$

where D_j is also a diagonal matrix. Then, for $D_j = \text{diag}(\theta_j^1, \dots, \theta_j^d)$, the matrix-valued Kuramoto model becomes juxtaposition of d Kuramoto models:

$$\dot{\theta}_j^p = e_p + \frac{\kappa}{N} \sum_{k=1}^N \sin(\theta_k^p - \theta_j^p)$$

- Thanks to Kuramoto literature, for $\kappa \gg 1$, there exists $\theta_j^{p,\infty}$ such that

$$\lim_{t \rightarrow \infty} \theta_j^p(t) = \theta_j^{p,\infty}.$$

- Similarly, we define

$$V_j(t) = e^{iD_j(t)} U_j,$$

and find its temporal evolution

$$\dot{V}_j = \frac{\kappa}{2N} \sum_{k=1}^N \left(e^{i(D_k - D_j)} (V_k - V_j) + (V_k - V_j) V_k^\dagger e^{i(D_j - D_k)} V_j \right)$$

- As done before, we can provide a sufficient condition leading to

$$\lim_{t \rightarrow \infty} V_j(t) = V^\infty.$$

Theorem (K.-Park, arXiv:2206.00984)

Let $\{U_j\}$ be a solution to (2) with $H_j = E_j$ where E_j is a diagonal matrix. If $\kappa \gg 1$ and $D(U_0) \ll 1$, then there exist a (constant) diagonal matrix D_j^∞ and $V^\infty \in U(d)$ such that

$$\lim_{t \rightarrow \infty} U_j(t) = U_j^\infty = e^{iD_j^\infty} V^\infty =: U_j^\infty.$$

High-dimensional Kuramoto model on sphere

- High-dimensional Kuramoto model on sphere has been proposed in several literature, e.g., [Olfati-Saber IEEE CDC '06, Lohe J. Phys. A '09, Zhu Phys. Lett. A '13, Tanaka New J. Phys. '14, etc.]

$$\dot{x}_j = \Omega_j x_j + \frac{\kappa}{N} \sum_{k=1}^N \left(x_k - \frac{\langle x_j, x_k \rangle}{\langle x_j, x_j \rangle} x_j \right) \quad (4)$$

- $x_j = x_j(t) \in \mathbb{R}^d$: position of j -th agent
 - Ω_j : $d \times d$ skew-symmetric matrix which plays a role of natural frequency
- Question: Does this model also exhibit emergent behavior similar to the Kuramoto model?

- As expected from Kuramoto literature, if Ω_i 's are identical, then one can show that relative distances between x_i converge to zero.

Theorem (Markdahl et al IEEE TAC '17, Ha-Ko-Ryo J. Stat. Phys. '18,)

Let $\{x_j\}$ be a solution to (4) with $\Omega_j \equiv \Omega$. Then, for generic initial data,

$$\lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0.$$

- Question: What happens for $\Omega_i \neq \Omega_j$ in general?

- In the Kuramoto world, if $\nu_i \neq \nu_j$, then complete synchronization cannot happen. Instead, phase-locked state can emerge for large κ .

Theorem (Ha-Ryoo Comm. Math. Phys. '20)

Let $\{\theta_j\}$ be a solution to (1). Then, there exists a critical coupling strength $\kappa_c > 0$ such that if $\kappa > \kappa_c$, then the following limit exists:

$$\lim_{t \rightarrow \infty} (\theta_i(t) - \theta_j(t)).$$

If Ω_i and Ω_j are distinct, we may expect complete synchronization cannot emerge.

Theorem (Chi-Choi-Ha J. Math. Phys '14)

Let $\{x_j\}$ be a solution to (4) with $\Omega_i \neq \Omega_j$ for some $i \neq j$. Then, if $\kappa \gg 1$ and $D(X_0) \ll 1$, then

$$\limsup_{t \rightarrow \infty} D(X(t)) \leq \mathcal{O}(\kappa^{-1}).$$

Here, $D(X(t))$ is the maximal diameter:

$$D(X(t)) := \max_{1 \leq i, j \leq N} \|x_i(t) - x_j(t)\|.$$

However, although Ω_i and Ω_j are different in general but $\{\Omega_i\}$ have a good structure: if there exists $n \in \mathbb{S}^{d-1}$ such that

$$\Omega_i n = \Omega_j n, \quad \text{for all } i, j \tag{5}$$

then complete synchronization can emerge.

Theorem (Zhu et al J. Stat. Phys. '22, Automatica '24)

Let $\{x_j\}$ be a solution to (4) with Ω_j satisfying (5). Then, if $\langle x_i, n \rangle > 0$, then we have

$$\lim_{t \rightarrow \infty} x_i(t) = n.$$

Question: Can the high-dimensional Kuramoto model can exhibit phase-locked state?

In other words, can we find some sufficient conditions leading to the state where the relative distances converge to some positive values?

For this, we assume

$$\Omega_i = \begin{pmatrix} 0 & D_i \\ -D_i & 0 \end{pmatrix} \quad (6)$$

Theorem (K., IEEE TAC '23)

Let $\{x_j\}$ be a solution to (4) with Ω_i satisfying (6). Then, if $\kappa \gg 1$ and $D(X_0) \ll 1$, then the limit of $\|x_i(t) - x_j(t)\|$ exists.

Here, we still do not know whether x_i converges or not.

We further assume

$$\Omega_i = \begin{pmatrix} 0 & a_i \\ -a_i & 0 \end{pmatrix}, \quad a_i \in \mathbb{R}. \quad (7)$$

By applying the same method,

Theorem

Let $\{x_j\}$ be a solution to (4) with Ω_j satisfying (7). Then, if $\kappa \gg 1$ and $D(X_0) \ll 1$, then there exists $x_j \in \mathbb{R}^{2d-1}$ such that

$$\lim_{t \rightarrow \infty} x_j(t) = x_j^\infty.$$

- ▶ We have studied the heterogeneous high-dimensional Kuramoto models by using the Kuramoto dynamics.
- ▶ For the high-dimensional Kuramoto model on unitary group, if we assume that the Hermitian matrix is just diagonal, we can show that a solution converges to equilibrium.
- ▶ Can we extend this result for general Hermitian matrix?

Happy Birthday!